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Assignment Sheet Nr. 1

Question 1: Logistic Regression — Theory

(a) The likelihood is the probability that a distribution will produce a certain dataset or point of data. Find an expression for the likelihood of a single data point i.e. $P(y = y^{(i)} \mid x^{(i)}, \theta)$ and the likelihood of the whole dataset i.e. $P(y \mid X, \theta) = \prod_i P(y = y^{(i)} \mid x^{(i)}, \theta)$. This expression should match with the one in part (b) when you take the logarithm.

$$
\begin{aligned}
P(y = y^{(i)} \mid x^{(i)}, \theta) &= \sigma(\theta^\top x^{(i)})^{y^{(i)}} (1 - \sigma(\theta^\top x^{(i)}))^{1-y^{(i)}} \\
P(y \mid X, \theta) &= \prod_i \left(\sigma(\theta^\top x^{(i)})^{y^{(i)}} (1 - \sigma(\theta^\top x^{(i)}))^{1-y^{(i)}} \right) \\
\log P(y \mid X, \theta) &= \sum_i \left(y^{(i)} \log \sigma(\theta^\top x^{(i)}) + (1 - y^{(i)}) \log\Big(1 - \sigma(\theta^\top x^{(i)})\Big) \right)
\end{aligned}
$$

(b) To find the optimal values of θ, we can maximize the likelihood by differentiating with respect to θ. However it is usually easier to work with the logarithm of the likelihood:

$$
\log P(y \mid X, \theta) = \sum_i y^{(i)} \log \sigma(\theta^\top x^{(i)}) + (1 - y^{(i)}) \log\Big(1 - \sigma(\theta^\top x^{(i)})\Big)
$$

Take the derivative of the log-likelihood and simplify it as far as possible.

$$
\begin{aligned}
\frac{d}{dz}\sigma(z) &= \sigma(z)(1 - \sigma(z)) \\
\frac{d}{dz}\log \sigma(z) &= \frac{\sigma'(z)}{\sigma(z)} = 1 - \sigma(z) \\
\frac{d}{dz}\log(1 - \sigma(z)) &= \frac{-\sigma'(z)}{1 - \sigma(z)} = -\sigma(z)
\end{aligned}
$$

$$
\begin{aligned}
\nabla_\theta \ell(\theta) &= \sum_{i=1}^{n} \left(y^{(i)}(1 - \sigma(z_i)) + (1 - y^{(i)})(-\sigma(z_i)) \right) x^{(i)} \\
&= \sum_{i=1}^{n} (y^{(i)} - \sigma(\theta^\top x^{(i)})) x^{(i)}
\end{aligned}
$$

(c) Now you should expand the logistic regression to a polynomial logistic regression of rank d, assuming x has only 2 elements. Therefore, you have to consider all combinations of elements of x, where the sum of their powers is $\leq d$. Derive an expression for $d = n$. How do you have to redesign θ and x?

To extend the model to degree $d = n$, we need all polynomial terms $x_1^i x_2^j$ so the sum of powers $i + j \leq n$. Then the sigmoid function becomes a sum over these combinations:

$$z = \sum_{0 \leq i+j \leq n} w_{i,j} x_1^i x_2^j$$

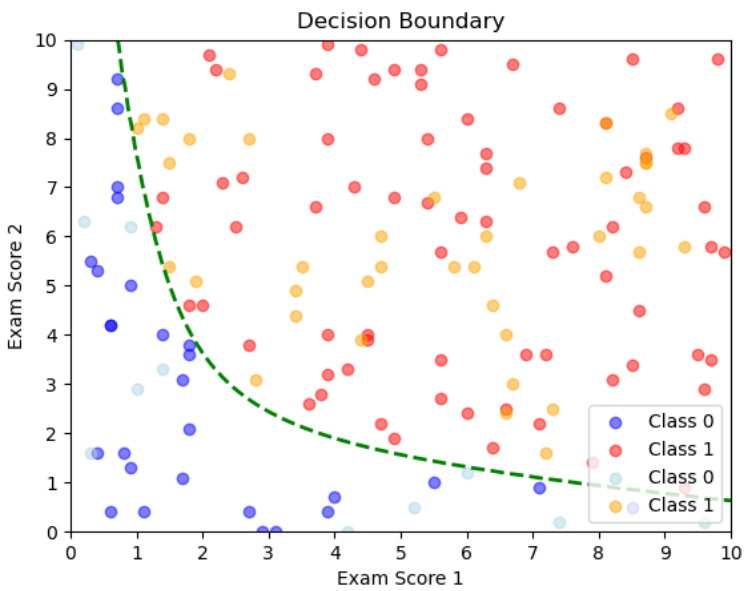
To have the vector notation $\theta^\top x$, we adjust x and θ as flattened vectors of dimension $D = \frac{(n+1)(n+2)}{2}$ which are these terms and their corresponding weights:

$$\boldsymbol{x}' = [1, x_1, x_2, x_1^2, x_1 x_2, x_2^2, \ldots, x_1^n, \ldots, x_2^n]^\top$$
$$\boldsymbol{\theta}' = [\theta_0, \theta_1, \theta_2, \ldots, \theta_{D-1}]^\top$$

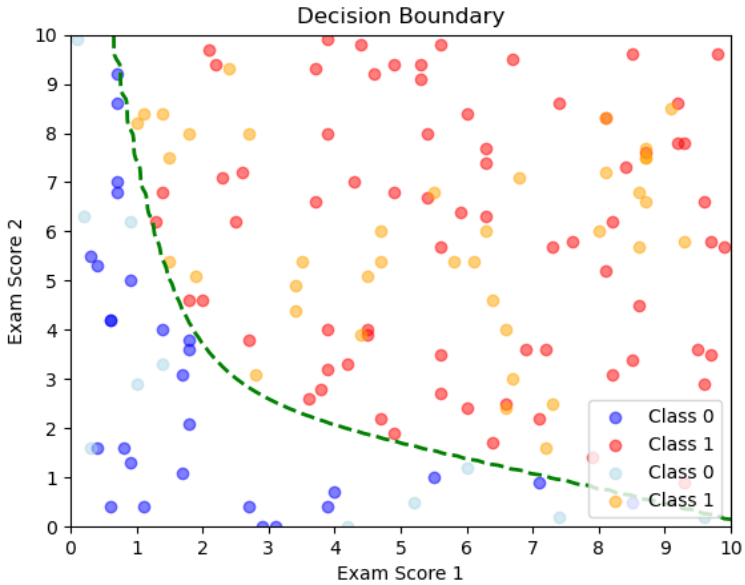
This then can then be written in the standard form $h(x) = \sigma(\boldsymbol{\theta}'^\top \boldsymbol{x}')$.

Question 2: Logistic Regression — Implementation

Decision Boundary for Polynomial Degree $d = 2$



Decision Boundary for Polynomial Degree $d = 3$



Results

While both the quadratic ($d = 2$) and cubic ($d = 3$) models achieved an accuracy of 1.0 on the test set, there are some differences in the training:

- Cost Function: The model with $d = 3$ achieved a lower final cost (0.0012) compared to the model with $d = 2$ (0.0125). This means that the cubic model fits the training data much more, pushing predicted probabilities closer to 0 and 1.
- Model Selection: Even though the lower cost of the cubic model, the quadratic model ($d = 2$) is most of the times the better choice. It achieves perfect classification with a simpler decision boundary thus reducing the risk of overfitting on future data.
- Stability: The $d = 3$ model had a much higher initial cost (5.41) compared to $d = 2$ (1.06). This suggests that without feature scaling, the higher-order terms (e.g., x^3) caused initial gradient instability.

Console Output

```
Step 0, Cost: 1.0637
Step 10000, Cost: 0.0368
Step 20000, Cost: 0.0263
Step 30000, Cost: 0.0216
Step 40000, Cost: 0.0188
Step 50000, Cost: 0.0168
Step 60000, Cost: 0.0153
Step 70000, Cost: 0.0142
Step 80000, Cost: 0.0133
Step 90000, Cost: 0.0125
Accuracy: 1.0
```

```
Step 0, Cost: 5.4118
Step 10000, Cost: 0.0051
Step 20000, Cost: 0.0037
Step 30000, Cost: 0.0028
Step 40000, Cost: 0.0023
Step 50000, Cost: 0.0020
Step 60000, Cost: 0.0017
Step 70000, Cost: 0.0015
Step 80000, Cost: 0.0014
Step 90000, Cost: 0.0012
Accuracy: 1.0
```